Vehicle Detector: A HOG+SVM and CNN based hybrid multi-vehicles detection system

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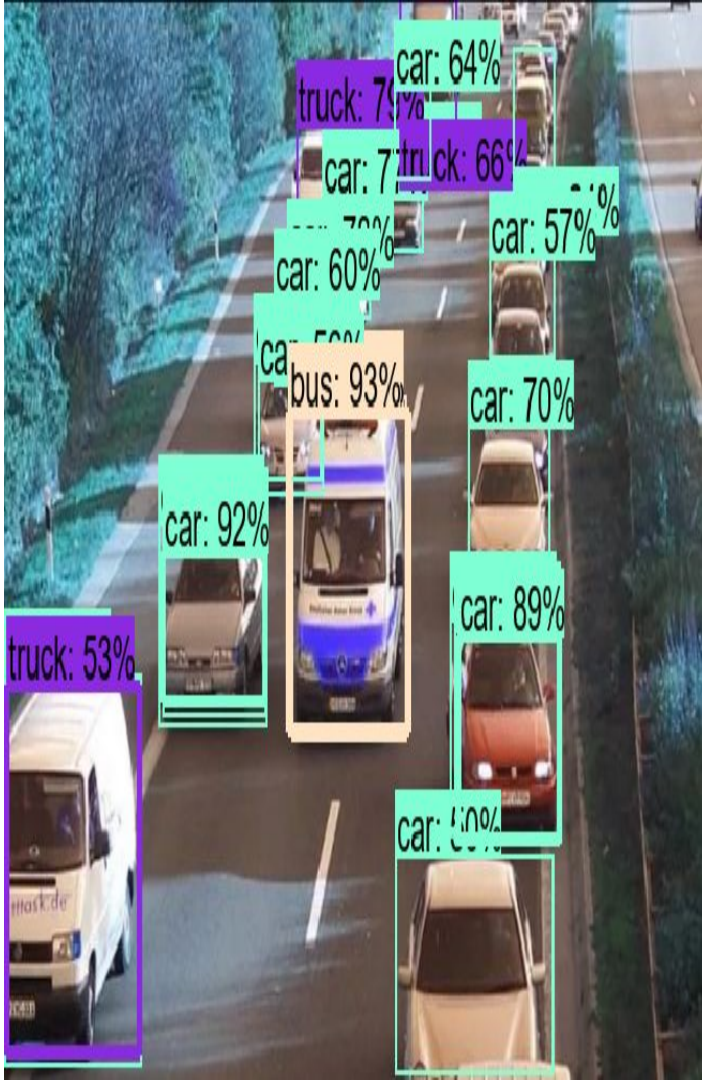
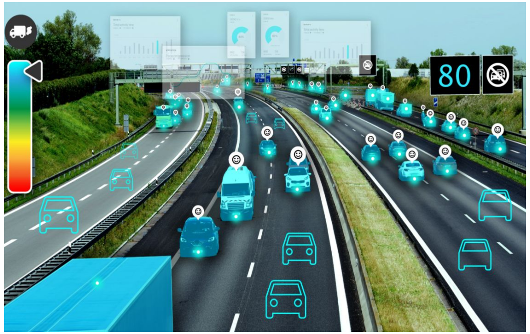
Course: COSC 444/544 - Computer Vision



Introduction

- **Vehicle detection is a crucial task in various applications**

Self-driving



Traffic monitoring

Security

- **Processing speed is critical in vehicle detection task**

Motivation & Objectives

Limitations of Traditional Methods:

- HOG+SVM: Sensitive to variations in lighting, viewpoint, and scale. High false positive rates.
- Haar Cascade Classifiers: Highly sensitive to occlusions and background clutter.
- SIFT:Computationally expensive

Objective:

- Find a solution that keeps a balance between accuracy and speed

Motivation & Objectives

Objectives:

- Easy to implement.
- High performance. Can be used in real-time scenarios.
- Low hardware requirements. Can be run in a embedded devices.

Our solution:

- Fuse classic traditional HOG+SVM method and deep network!

Related Work

Traditional Methods: Pros and Cons

Advantages

Interpretability:

- Features like HOG, SIFT, and SURF offer clear insights into what aspects of an image are being used for detection. [5, 11, 3]

Computational Efficiency:

- Techniques such as Haar cascades and SVM-based methods provide fast processing, suitable for real-time applications in low-resource environments. [16]

Limitations

Sensitivity to Environmental Variations:

- Methods can struggle with occlusion, varying lighting conditions, and diverse viewpoints.

Handcrafted Feature Dependency:

- The need for manually designed features may limit robustness across diverse scenarios.

Scalability Challenges:

- While efficient for binary classification (vehicle vs. non-vehicle), extending these methods to multi-class detection (e.g., differentiating between sedans, SUVs, trucks) is non-trivial. [8, 13]

Related Work

Deep Learning Approaches: Performance vs. Cost

High Accuracy

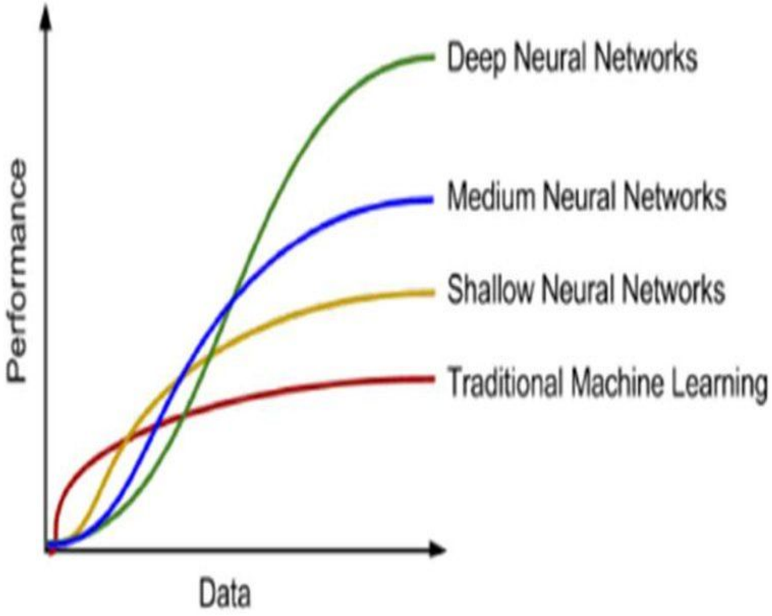
Models like Faster R-CNN, SSD, DETR, and YOLO achieve state-of-the-art performance even in challenging conditions. [4, 17, 14, 15]

Computational Cost

These methods often demand extensive labeled datasets and high computational power

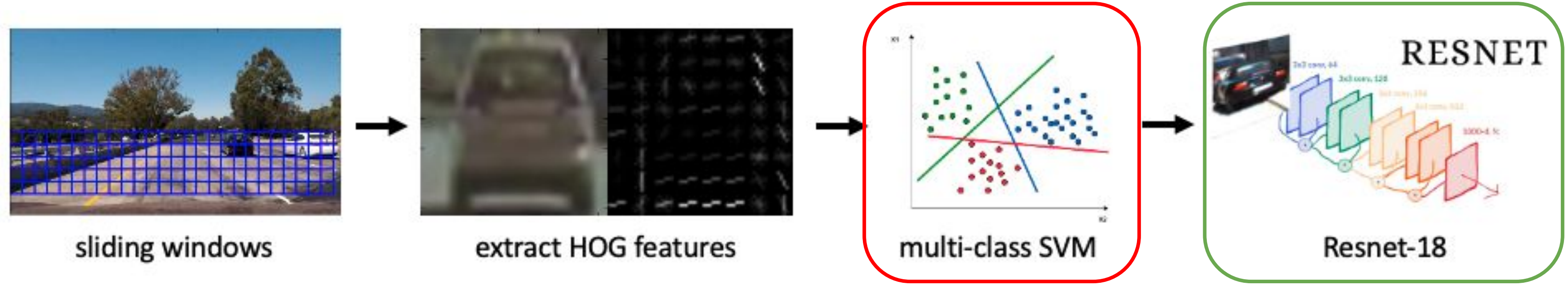
Interpretability

Unlike traditional approaches, the features learned by deep networks can be less interpretable.



Our Proposed System

Our hybrid vehicle detection pipeline



Preliminary detection

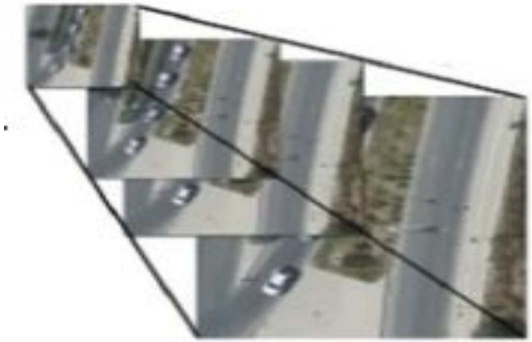
Consolidation

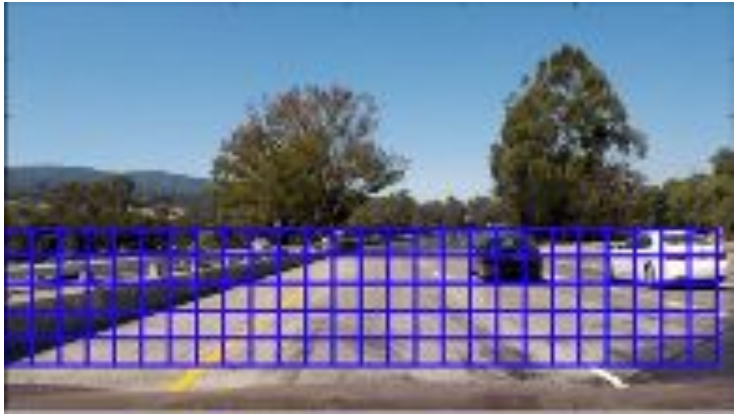
Input: A single image that could contain multiple vehicles

Output: Image patches and vehicle labels

Our Proposed System

Sliding window



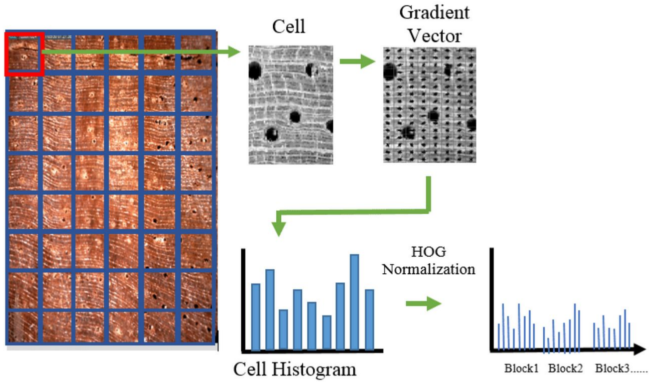


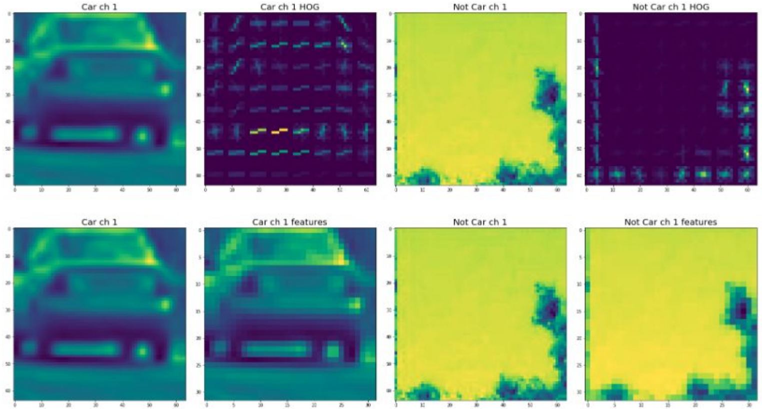
Create a image pyramid with 7 scales for each input image

Divide each scale into 16x16 pixels block and comput HOG on a 64x64 patch

Our Proposed System

Extract HOG features



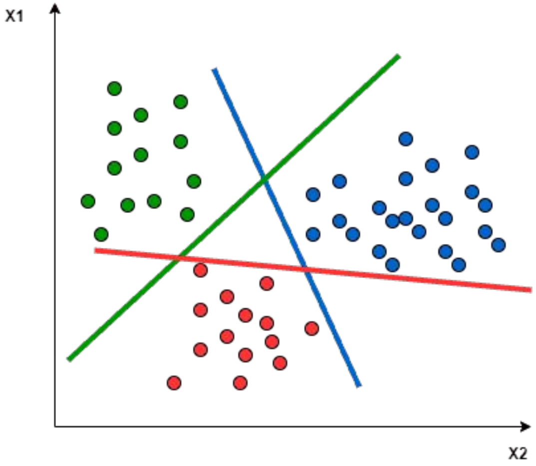


Use scikit-image to compute HOG features

Block size 2x2 cells, cell size 16x16 pixels, feature size 2052

Our Proposed System

Multi-class SVM



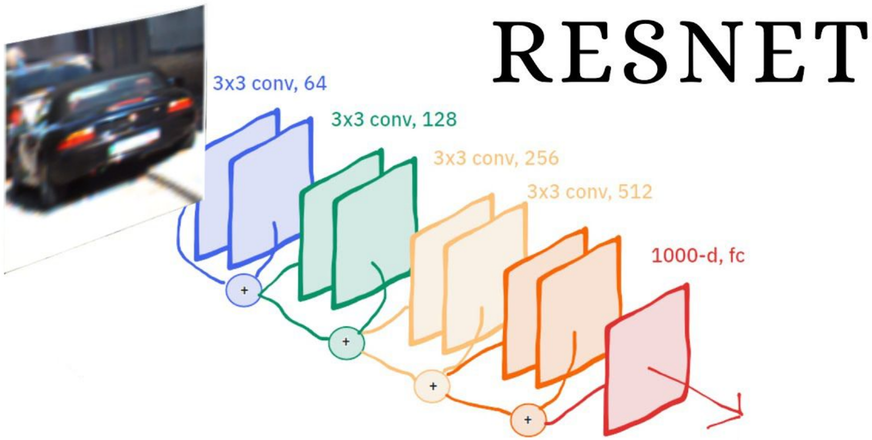
$$softmax(\mathbf{z})_i = \frac{e^{z_i}}{\sum_{j=1}^{N} e^{z_j}}$$

- Use multi-class SVM in scikit-learn library in OvO strategy
- Use SoftMax to convert distances to hyperplane to a confidence
- Accept HOG+SVM’s result if confidence > 0.8

Our Proposed System

Resnet-18 CNN

- **Less computation requirement**
- **Parameter number 11689512, model size 44MB**
- **A single inference takes 0.03ms**
- **Use pre-trained weight, fine-tuning in out dataset**



Dataset

Our dataset is collected from multiple public datasets

Roboflow

Images.cv

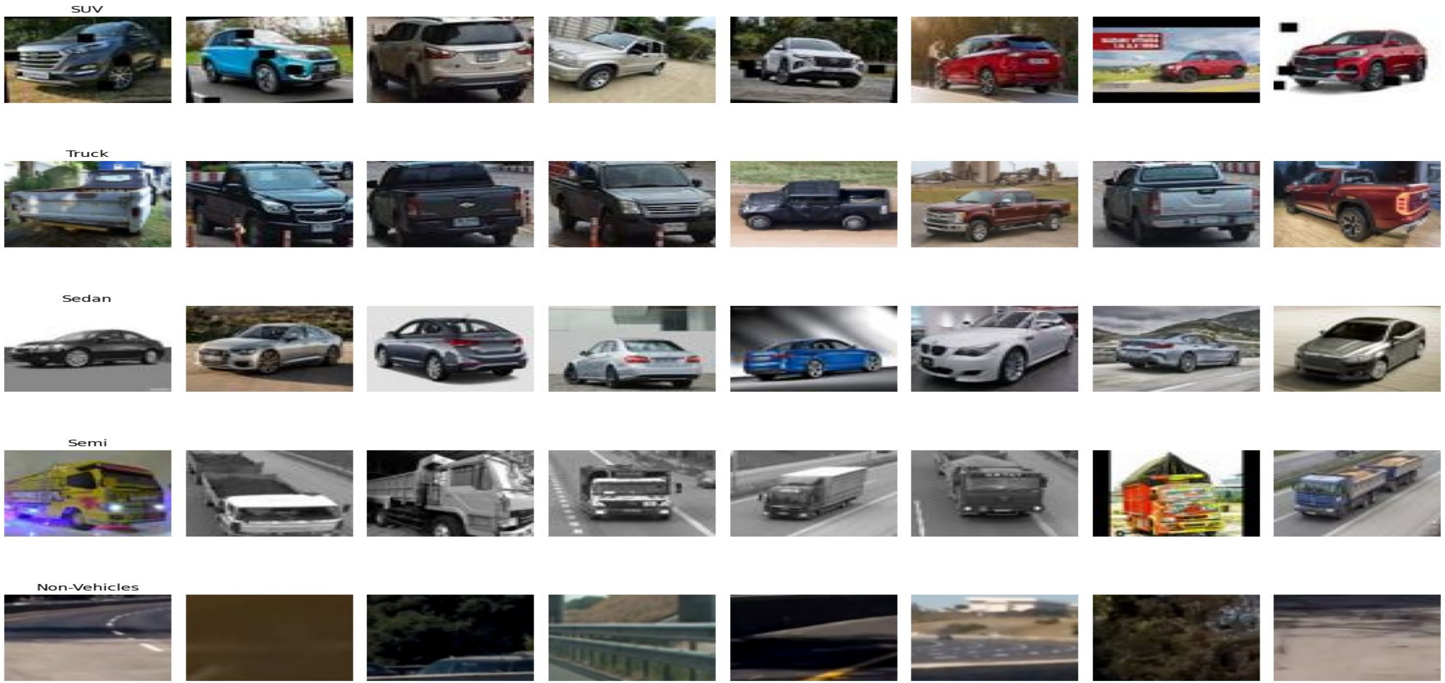
Mendeley Data

Udacity

Dataset distribution (5 categories)

- Sedan: 1000 samples (10.9%)
- Truck: 1107 samples (12.1%)
- SUV: 1200 samples (13.1%)
- Semi: 1871 samples (20.5%)
- Non-Vehicles: 3968 samples (43.4%)

Dataset



Example

Data Augmentation

Result

HOG+SVM

training time: took 112s for a dataset with 9146 images on a CPU

inference speed: ~0.9ms for a single patch, test performed on 50 image in size 64x64 pixels

Resnet-18 CNN

training detail: dropout 0.5, lr 0.001, Adam optimizer, epoch 20, batch size 32

training time: took 25min for a dataset with 9146 images on a TeslaA100

inference speed: ~30ms for a single patch , test performed on 50 image in size 64x64 pixels

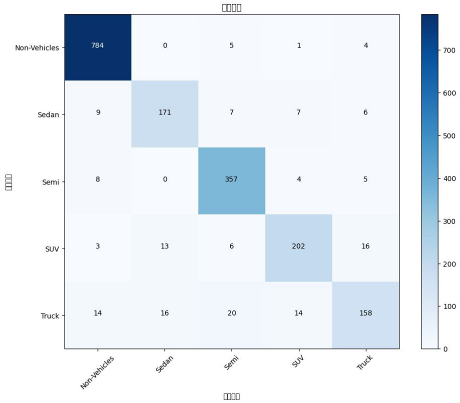
HOG+SVM is 30x faster than CNN!

Result

HOG+SVM

Overall accuracy = 94.26%

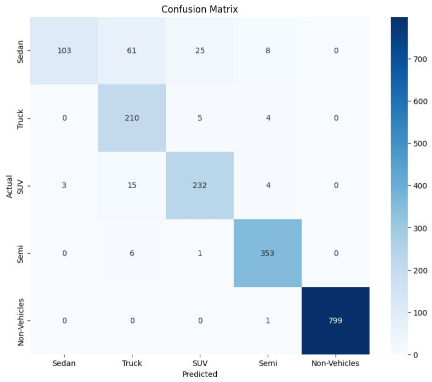
	precision	recall	f1-score	support
Non-Vehicles	0.96	0.99	0.97	794
Sedan	0.85	0.85	0.85	200
Semi	0.90	0.95	0.93	374
SUV	0.89	0.84	0.86	240
Truck	0.84	0.71	0.77	222



Resnet-18 CNN

Overall accuracy = 96.78%

	precision	recall	f1-score	support
Sedan	0.97	0.52	0.68	197
Truck	0.72	0.96	0.82	219
SUV	0.88	0.91	0.90	254
Semi	0.95	0.98	0.97	360
Non-Vehicles	1.00	1.00	1.00	800

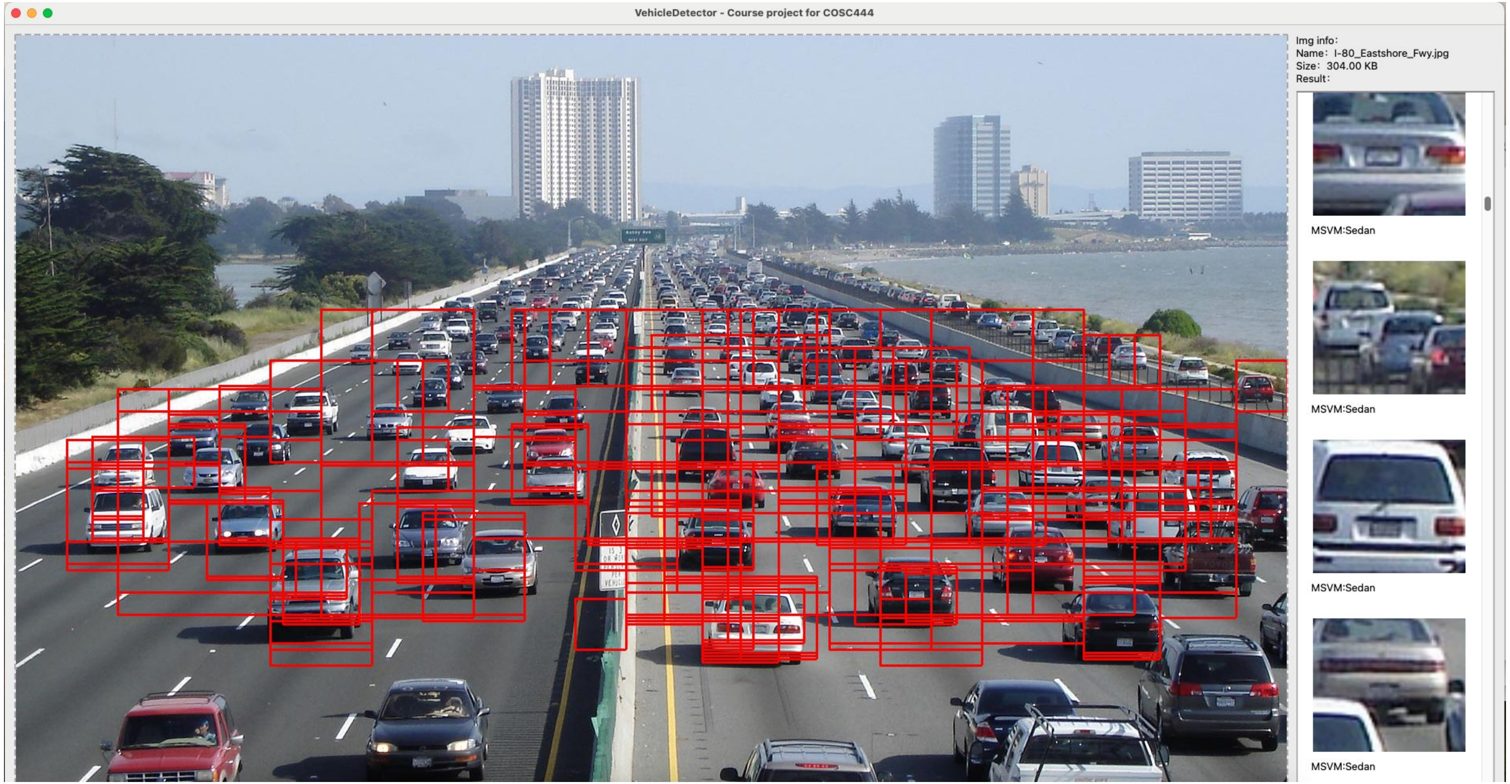


Result

Here CNN corrected the result!

System user interface

Result



Result of an image with many vehicles

Conclusion

- We present a **hybrid** vehicle detection system combining **HOG+SVM** with a **ResNet-18 CNN**.
- We use **sliding windows** for initial region proposal.
- **HOG+SVM** is used for **initial vehicle detection** and **classification**.
- A **ResNet-18 CNN** is applied for **prediction consolidation**.
- The hybrid architecture leverages HOG's efficiency and CNN's advanced feature learning for robust and accurate real-time vehicle detection.
- Our system might be used in real-time scenarios

Thank you
Any questions?

References

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[12] Zhao & Nevatia. 2003. *Car Detection in Low Resolution Aerial Images.*

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[17] Erhan et al. 2016. SSD: Single Shot MultiBox Detector.